

Model Evaluation Summary — Milestone 3

Project: BTC Volatility Spike Detector
Date: 2026-04-05

Evaluation Setup

Parameter	Value
Target	<code>vol_spike</code> — 1 if 60-second realized volatility $\geq \tau = 0.000046$
Target horizon	60 seconds
Features	<code>log_return</code> , <code>spread_bps</code> , <code>vol_60s</code> , <code>mean_return_60s</code> , <code>trade_intensity_60s</code> , <code>n_ticks_60s</code>
Validation strategy	Time-based splits with no shuffling
Split	60% Train / 20% Validation / 20% Test (by row count)
Primary metric	PR-AUC (precision-recall area under curve)

Dataset Context & Regime Shift

The dataset spans approximately 18 hours of continuous BTC-USD tick data (2026-04-04 22:54 → 2026-04-05 17:10). Something to notice is that the overnight session was significantly calmer than the morning session that followed.

Split	Rows	Spike Rate	Period
Train	66,578	4.3%	Overnight (low-volatility regime)
Validation	22,193	19.9%	Morning session open
Test	22,193	16.3%	Morning session continuation

The spike rate jumped from 4.3% in training to ~16–20% in evaluation. This **4x increase** is driven entirely by real-world intraday trading patterns. This is consistent with the Evidently drift report, which detected statistically significant drift in between the training and test windows.

To handle the rare positive class in trainin, the Logistic Regression model used `class_weight="balanced"`, which up-weights the 4.3% spike minority during training so the model still learns to detect them.

Results

Model	Val PR-AUC	Test PR-AUC	Val F1 (best)	Test F1 (best)
Z-score baseline (<code>z > 2.0</code> on <code>vol_60s</code>)	0.3341	0.2611	0.4313	0.3567
Logistic Regression (<code>C=0.1</code> , balanced)	0.2523	0.4163	0.3534	0.4195

Baseline parameters: `zscore_threshold = 2.0` , `feature = vol_60s` , calibrated on train mean/std.
LR parameters: `C = 0.1` , `solver = lbfgs` , `class_weight = balanced` , `tau = 0.3` .

Conclusion

The Logistic Regression model **outperforms the z-score baseline on unseen test data** (PR-AUC 0.4163 vs 0.2611), demonstrating that the learned combination of features performs better than using a single feature.

However, both models face a genuine challenge. They were trained on a 4.3% spike rate and evaluated showcasing a 16–19% spike rate. Collecting data across multiple full days could help stabilize the training distribution and possibly improve both PR-AUC scores.